AMEER SALMAN M

Forward Deployed Engineer, Databricks — data platforms, streaming pipelines, regulated payments

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SUMMARY

Forward Deployed Engineer at Databricks, embedding with enterprise customers to ship Data and AI solutions end to end in their own environments. Delivered **$180K/year in verified cost savings**, contributed an algebraic migration-validation engine into Databricks' open-source **Lakebridge** (**8.1×** faster, **~48,000×** less data moved), and turned a single customer's Spark connector blocker into maintained field-wide tooling covering six Databricks runtimes. Previously seven years at Amazon Pay building the pipelines behind India's regulated payment reconciliation, exiting as Data Engineer II — cutting reporting SLA from 36–40 hours to 30 minutes and reconciling ~400M transactions/month under a statutory five-working-day deadline.

EXPERIENCE

Forward Deployed Engineer · Databricks

JUL 2024 — PRESENT

INDIA | CURRENT ROLE

- Led optimisation of **4–5 of the top-20 most expensive Databricks jobs** for an enterprise customer via query rewrites, partitioning and shuffle reduction — **$180,000/yr verified savings** ($132K/50%, $43K/30%, $5K/10%) with zero change to business output; 2 further jobs in verification.
- Designed and built **DataPrint**, an algebraic data-fingerprinting engine validating billion-row migrations via JDBC-pushdown aggregates instead of moving data — **8.1×** faster on real 74M-row Redshift data, **~48,000×** less transferred, validated to **15.8B rows × 28 columns** with **zero false positives/negatives** and **≥98.8%** exact-difference solve rate.
- Contributed DataPrint into Databricks' open-source **Lakebridge** migration tool behind an opt-in, fail-open flag; fixed 3 latent Redshift↔Databricks serialisation bugs that were silently corrupting every Redshift reconciliation.
- Resolved a **Spark SQL Server connector** break blocking an enterprise customer's **Unity Catalog migration** after Microsoft archived the project — custom schema resolver for **Spark 3.5 and 4.0** plus multi-profile Maven builds for **Scala 2.12/2.13**. Only known workaround; adopted organically across multiple accounts.
- Productised it into a **maintained shared fork** via a ~250-line public-API delegation layer with no per-version branching — one codebase covering **Databricks Runtime 14.3–18.x** across 3 build profiles, retaining **~15×** write throughput vs generic JDBC; CI matrix with weekly runtime smoke tests, **testcontainers** integration and a per-release 1M-row perf-regression gate.
- Built **ShadowClone**, a production-risk-free optimisation sandbox on **Delta shallow clones**, **Unity Catalog lineage** and system tables with a fail-closed identity boundary and bidirectional `EXCEPT ALL` parity — proved a **301.5→29.6 min (10.19×)** optimisation at lower DBU with **21 tables identical across 4.45B rows**, inherited by **30+ pipeline families**.
- Built **AgentGuard**, a runtime security gateway intercepting, risk-scoring, auditing and rolling back AI-agent actions before production via a **6-checkpoint pipeline** — backed by a tamper-evident append-only audit ledger and three tiers of rollback on **Delta Time Travel**; built against **EU AI Act Article 14** oversight and audit requirements.
- Built **Infrastructure and Security as Code** in **Terraform** across test/stage/prod — reusable modules for cluster policies, schema creation and Unity Catalog grants; resolved a circular IAM trust dependency and a Terraform state mismatch; **~15 resource templates** turning per-team provisioning into a repeatable process; onboarded 4 new catalogs.
- Architected a centralised migration platform (discovery → delivery) with **GitHub Actions CI/CD** for Databricks Apps and led its front end from **Streamlit to React/FastAPI**; unblocked a failing **Lakeflow Connect** ingestion pipeline by isolating column-masking, DDL-capture and Azure SQL privilege faults across three teams.

Data Engineer II · Amazon Pay

OCT 2017 — JUL 2024

BENGALURU & CHENNAI | 6 YRS 10 MOS | EXIT TITLE

- Rebuilt batch order-transaction reporting as event-driven streaming ingestion using database change streams, cross-account S3 replication and Lambda — cutting downstream SLA from **36–40 hours to 30 minutes** (~99%), live **8 months with zero defects**.
- Built a near real-time merchant metrics pipeline across accounts and regions with customer-managed KMS encryption, partitioned Parquet, Glue crawlers, Athena and Redshift — **15–30 minute SLA**, **100% accuracy**, **zero job failures**, with restricted PII excluded in flight before durable write.
- Automated a **20 million record** negative-remittance load with Python and Boto3 including pre-commit reconciliation — 100% accuracy, ~2 days of manual effort removed, clearing a multi-billion-rupee exception backlog.

- Orchestrated **~60 report jobs and 30 dashboards** into three one-click workflows and automated variance root-cause analysis (~4 days → automated), holding the **5-working-day** regulatory substantiation deadline and moving substantiation from monthly to daily.
- Introduced pipeline-as-code CI/CD across two data platforms with **dual sign-off SQL review** (business logic and query efficiency), eliminating drift between deployed jobs and source control.
- Built a self-mutating **AWS CDK** deployment pipeline spanning Beta, Gamma and Prod across multiple AWS accounts and regions, with approval workflow, security scanning, integration tests and bake time.
- Ran reconciliation across **~15 source systems and ~120 ETL jobs**, categorising **~400 million transactions/month** into daily and monthly transaction-level reports; reduced manual ETL workload 60%.
- Led migration of Redshift, QuickSight and S3 to the Mumbai region for **RBI payment aggregator data localisation**; supported RBI audit root-cause analysis across India regulatory accounts.
- Reclaimed **20% of Redshift disk** and eliminated CPU-driven cluster outages by replacing cluster-wide vacuum with targeted ANALYZE and deep-copy on high-impact tables.
- As Automation Specialist, built Python and Selenium automation for campaign management and seller onboarding, saving **~15 FTE** of effort; authored SOPs enabling stakeholder self-service.

TECHNICAL SKILLS

LANGUAGES	Python, SQL, Scala, Java, TypeScript, Bash	LAKEHOUSE	Databricks, Delta Lake, Unity Catalog, Photon, Asset Bundles, Lakeflow Connect
PROCESSING	Apache Spark (internals, 3.5/4.0), PySpark, AWS Glue, EMR, Parquet	INGEST	DynamoDB Streams, Kinesis, S3 replication, Lambda, JDBC / CDC
WAREHOUSE	Redshift, Athena, SQL Server, Postgres, MySQL, DynamoDB	PLATFORM / IAC	Terraform, AWS CDK, CloudFormation, Managed Airflow, GitHub Actions, Maven
TESTING	Pytest, ScalaTest, testcontainers, property-based testing, CI perf gates	FULL STACK	FastAPI, React/TypeScript, Streamlit, Databricks Apps
SECURITY	IAM, KMS, OAuth, secrets management, audit ledgers	GOVERNANCE	Unity Catalog, GDPR, PII exclusion, EU AI Act Art. 14, RBI localisation

SELECTED PROJECTS & OPEN SOURCE

- **DataPrint → Databricks Lakebridge** — algebraic migration-validation engine, contributed upstream. 8.1× faster, ~48,000× less data moved, validated to 15.8B rows.
- **Spark MSSQL Connector** — public-API compatibility layer covering Databricks Runtime 14.3–18.x from one codebase; ~15× write throughput vs generic JDBC.
- **ShadowClone** — production-risk-free optimisation sandbox; proved 10.19× with byte-level parity across 4.45B rows.
- **AgentGuard / LakeGuard** — runtime security gateway for AI agents; 6-checkpoint pipeline, hash-chained audit ledger, Saga-pattern rollback.
- **AWS automation toolkit** — Python tooling for EC2 snapshot lifecycle management. github.com/ameerafi

EDUCATION

DEGREE	B.E. Mechanical Engineering — Park College of Engineering & Technology, Coimbatore, 2017